

Group 7

Smart Door

ประตูอัจฉริยะ

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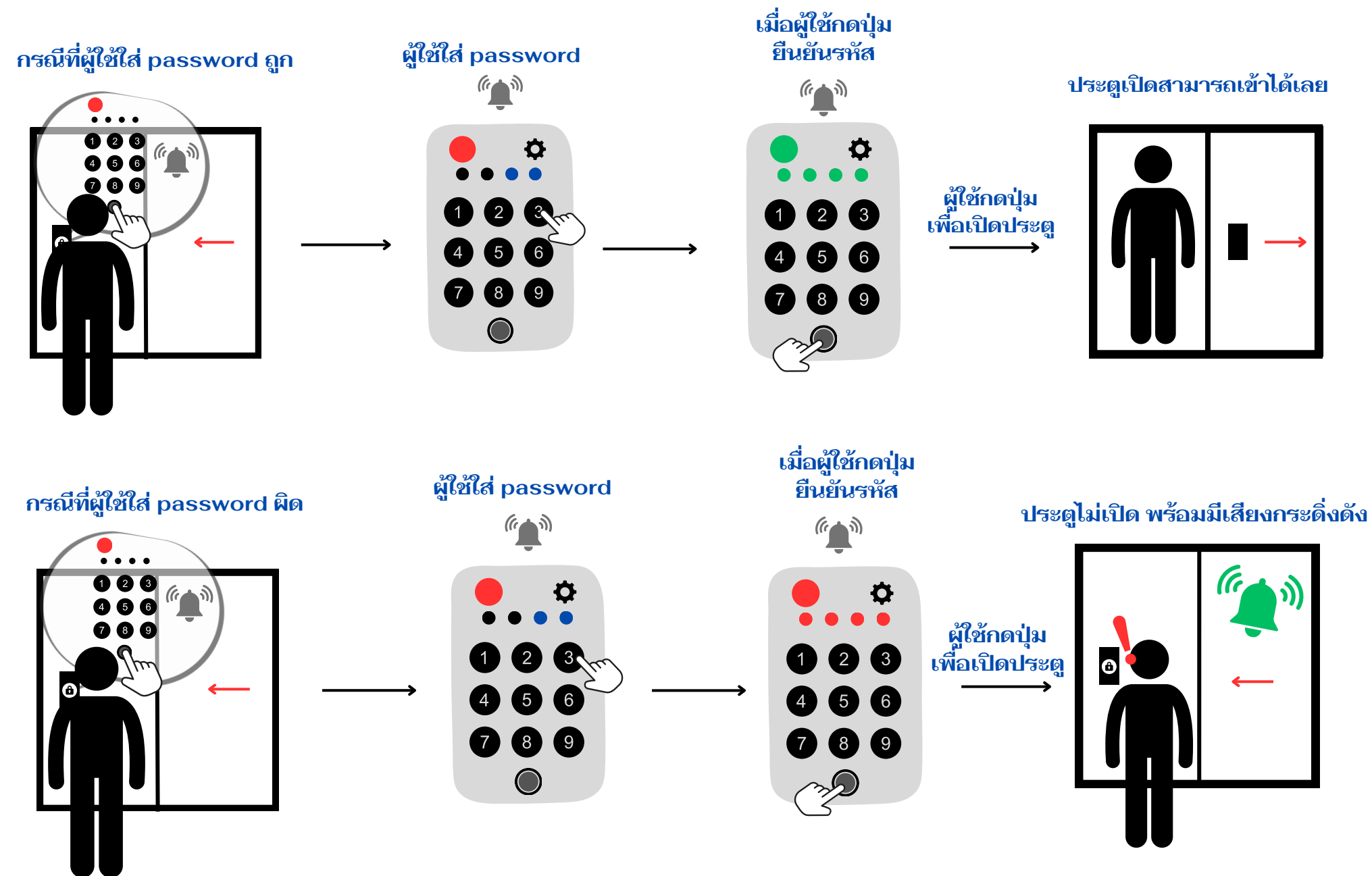


Scenario Diagram

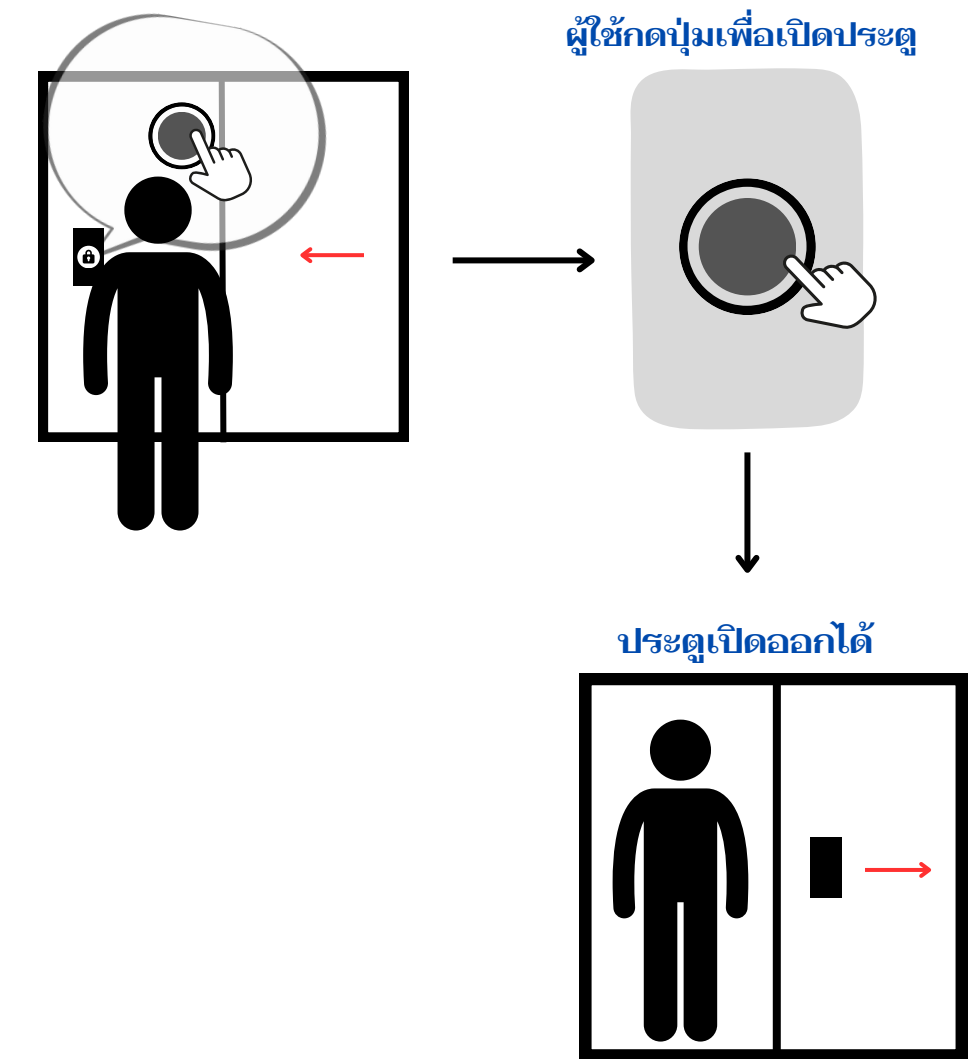
ปัญหา : ไม่สามารถตรวจสอบได้ว่าใครมีสิทธิ์ที่จะเข้าบ้านได้บ้าง

วิธีแก้ : ทำระบบ lock / unlock mode และกรอก password ให้ถูกต้องเพื่อเปิดประตู

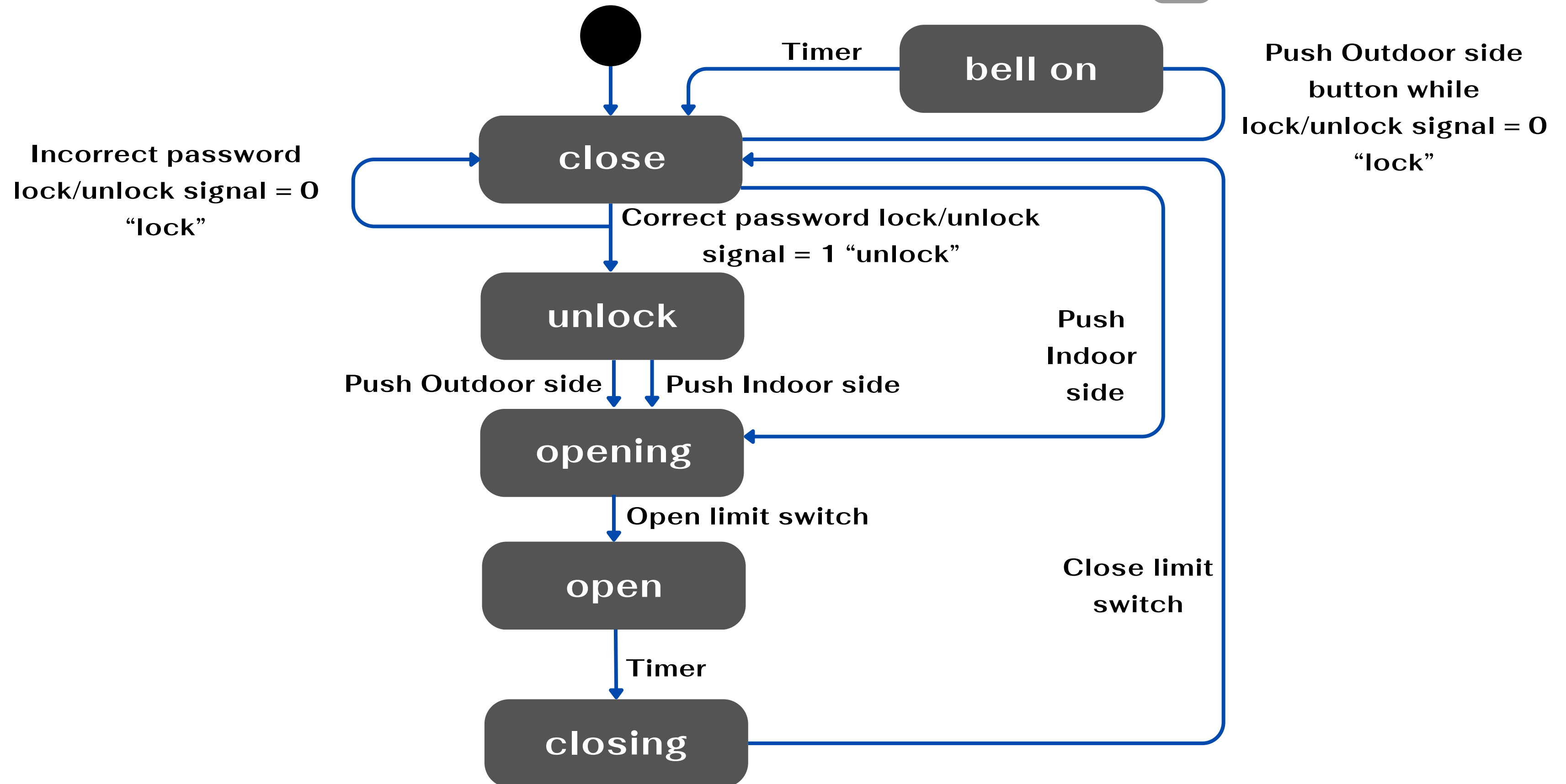
Outdoor



Indoor



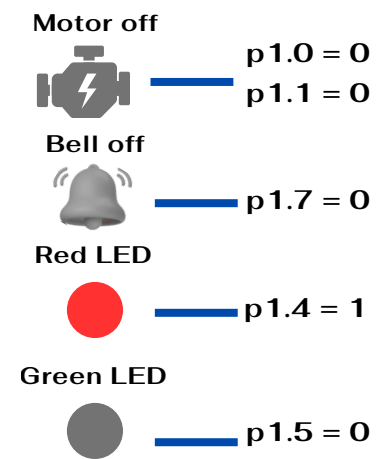
State Machine Diagram



State Description

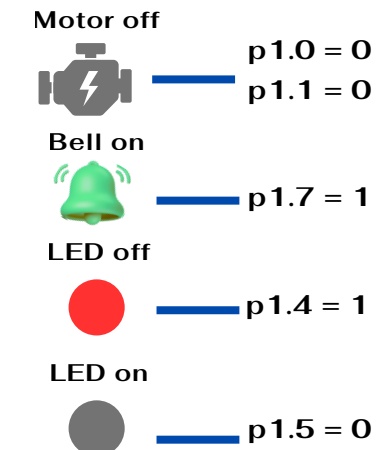
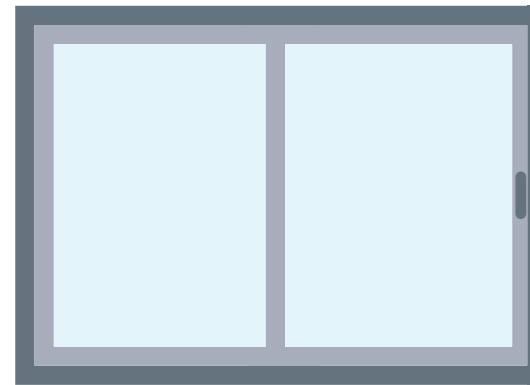
1.Close State

The door is closed.



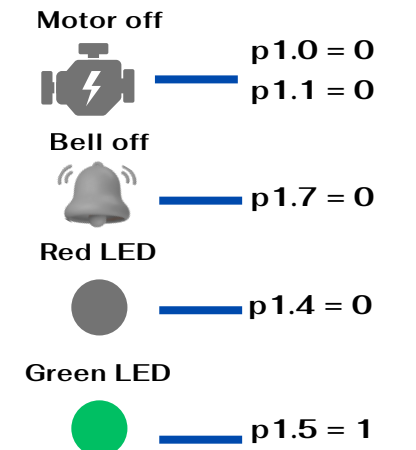
2.Bell On State

The door is closed.



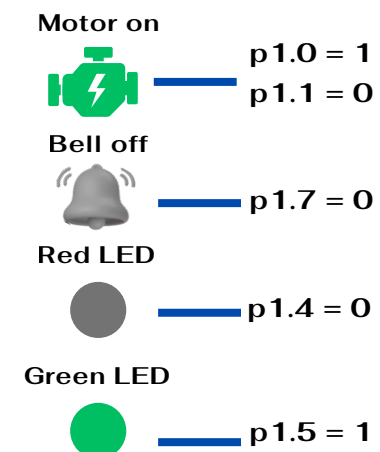
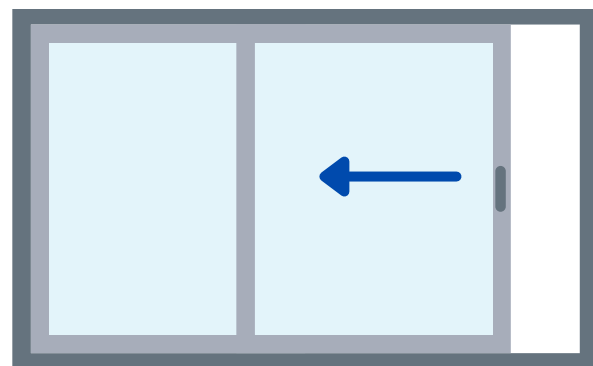
3.Unlock State

The door is closed.



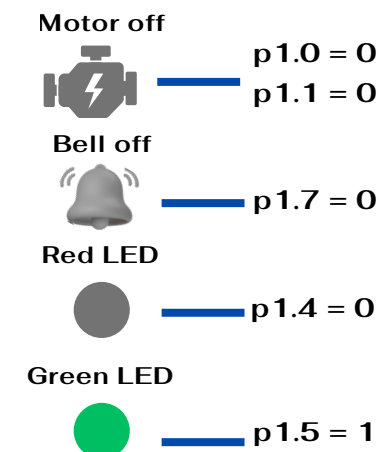
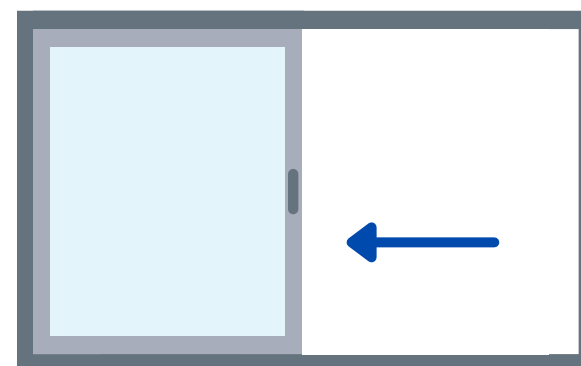
4.Opening State

The door is opening.



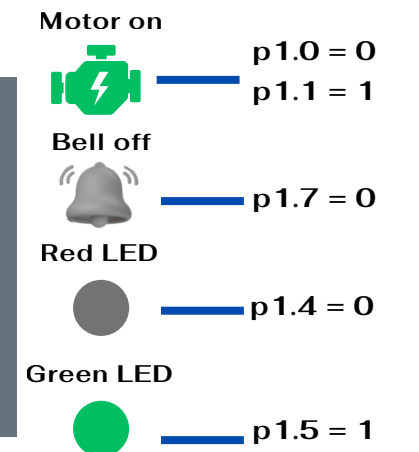
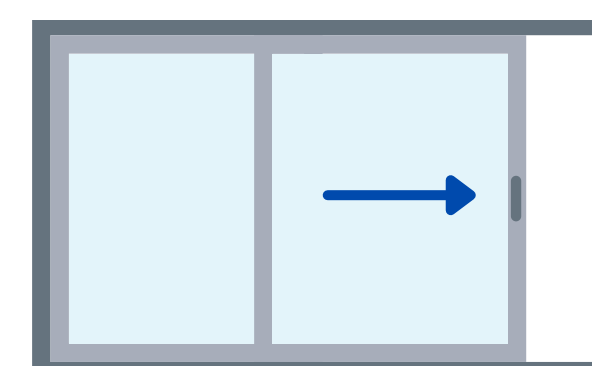
5.Open State

The door is open.

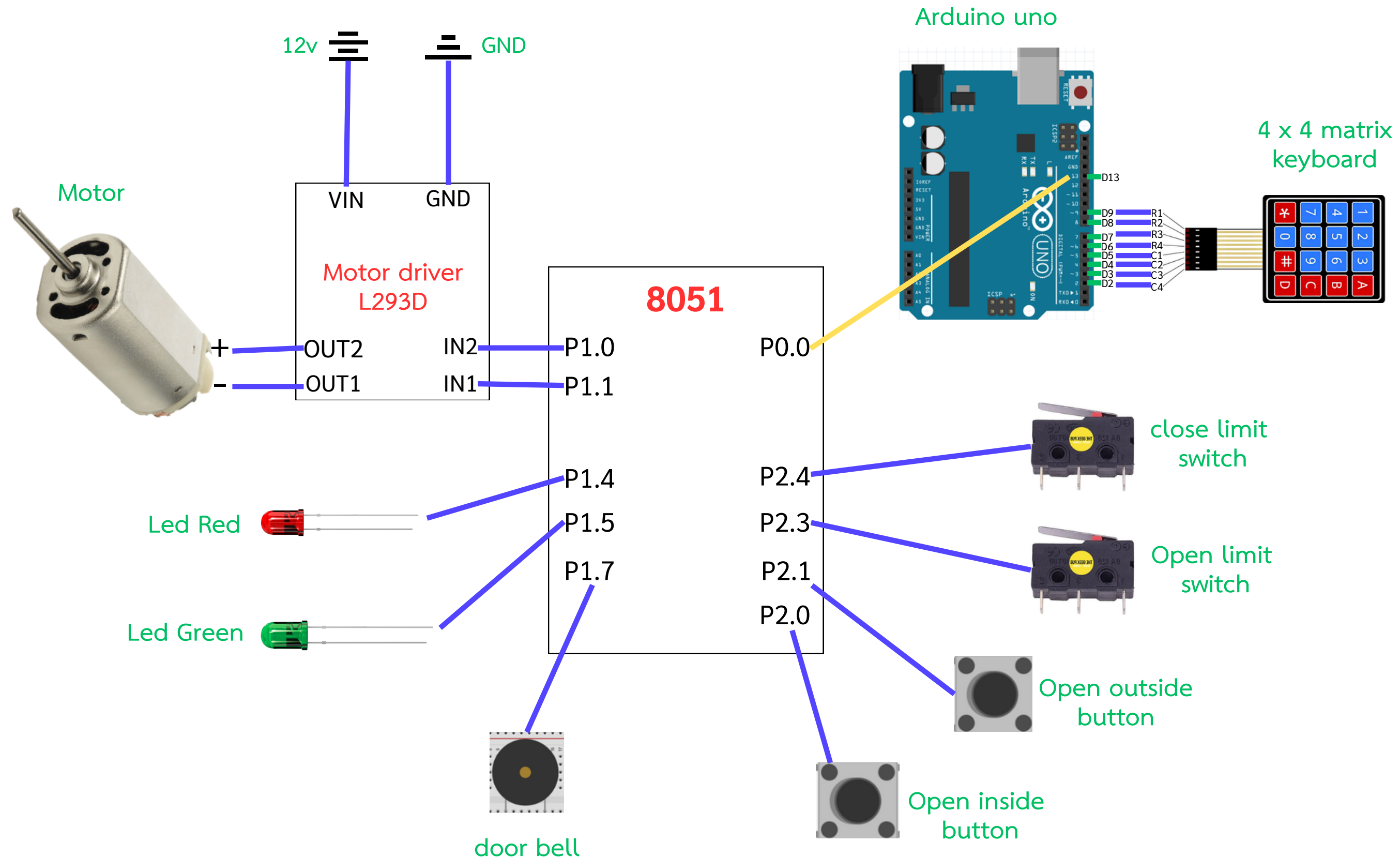


6.Closing State

The door is closing.

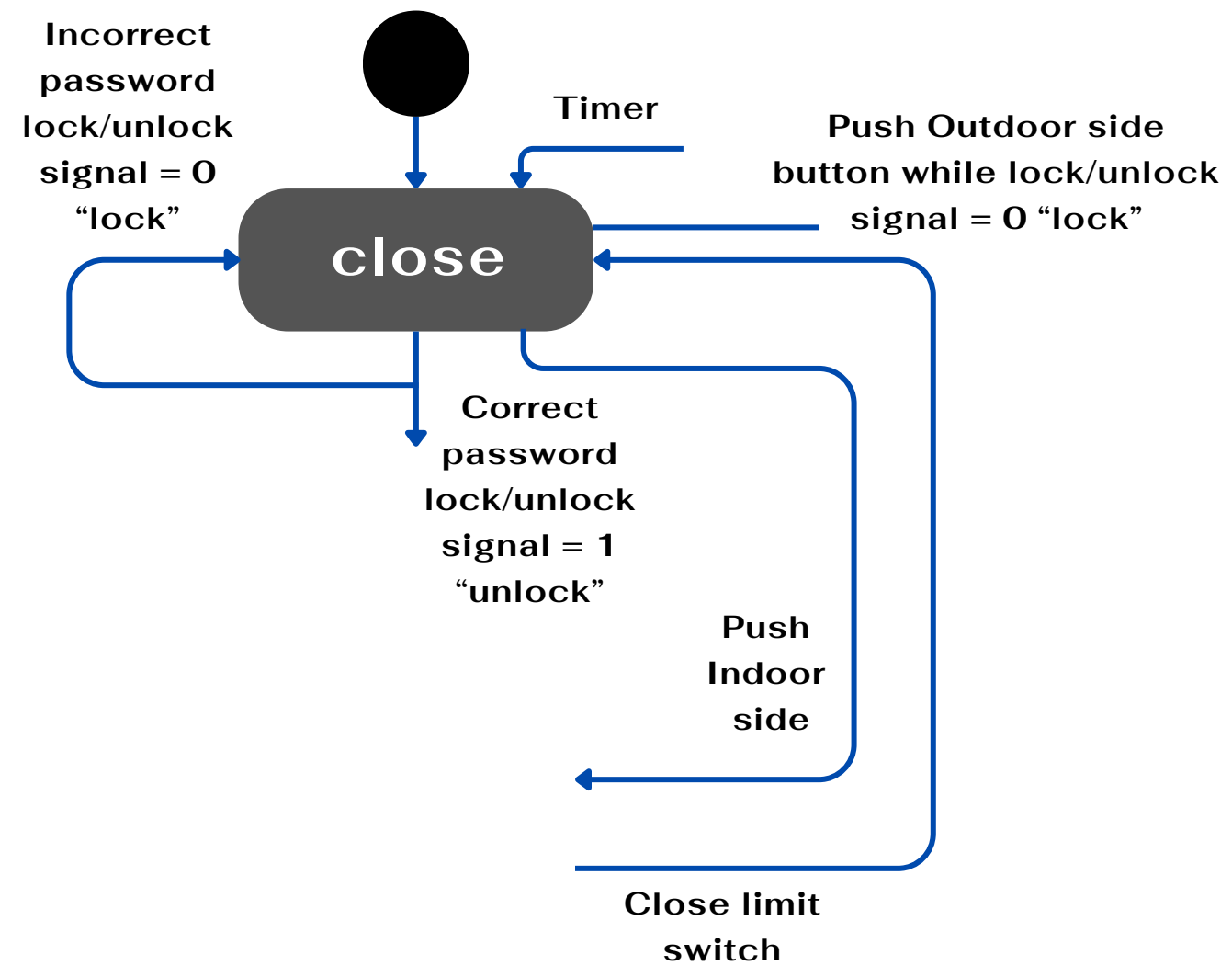


Hardware Diagram



Code

Close State



```
; p1.0 ; + pole of motor
; p1.1 ; - pole of motor

; p1.4 ;led red
; p1.5 ;led green

; p1.7 ;bell

; p0.0 ;Unlock Signal

; p2.0 ;open inside button
; p2.1 ;open outside button

; p2.3 ;open limit switch
; p2.4 ;close limit switch
```

```
CloseState:
mov r0,#01H ;r0 = state 1

clr p1.0 ; + pole of motor Off
clr p1.1 ; - pole of motor Off

setb p1.4 ;led red On
clr p1.5 ;led green Off

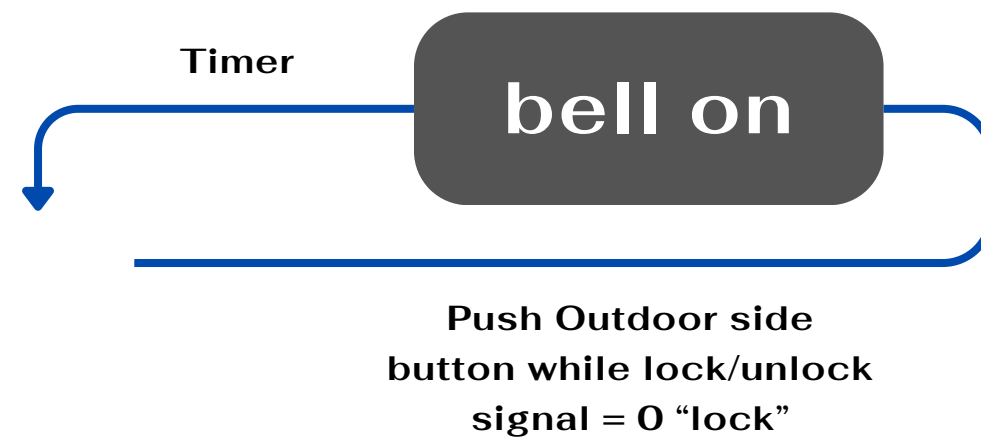
clr p1.7 ;bell Off

jB p2.0,OpeningState;open inside button
jB p2.1,BellOnState;open outside button
jB p0.0,UnlockState ;unlock Signal

sjmp CloseState
```

Code

Bell On State



BellOnState:

```
setb p1.7 ;bell On
```

```
mov r1,#25 ;delay 1.6 sec
```

```
timerC:
```

```
call delay
```

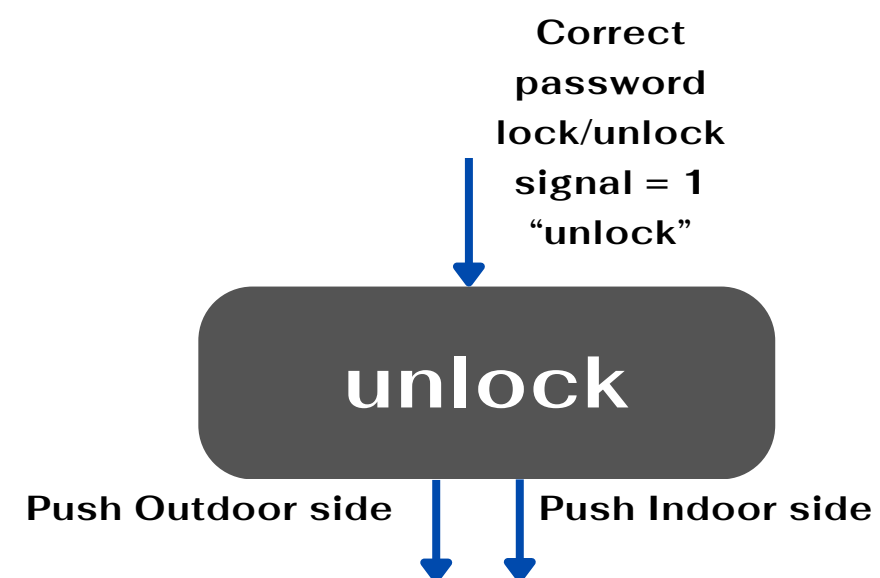
```
djnz r1,timerC
```

```
clr p1.7 ;bell Off
```

```
sjmp CloseState
```

Code

Unlock State



UnlockState:

```
mov r0,#02H ;r0 = state 2
```

```
clr p1.0 ; + pole of motor Off
```

```
clr p1.1 ; - pole of motor Off
```

```
clr p1.4 ;led red Off
```

```
setb p1.5 ;led green On
```

```
clr p1.7 ;bell Off
```

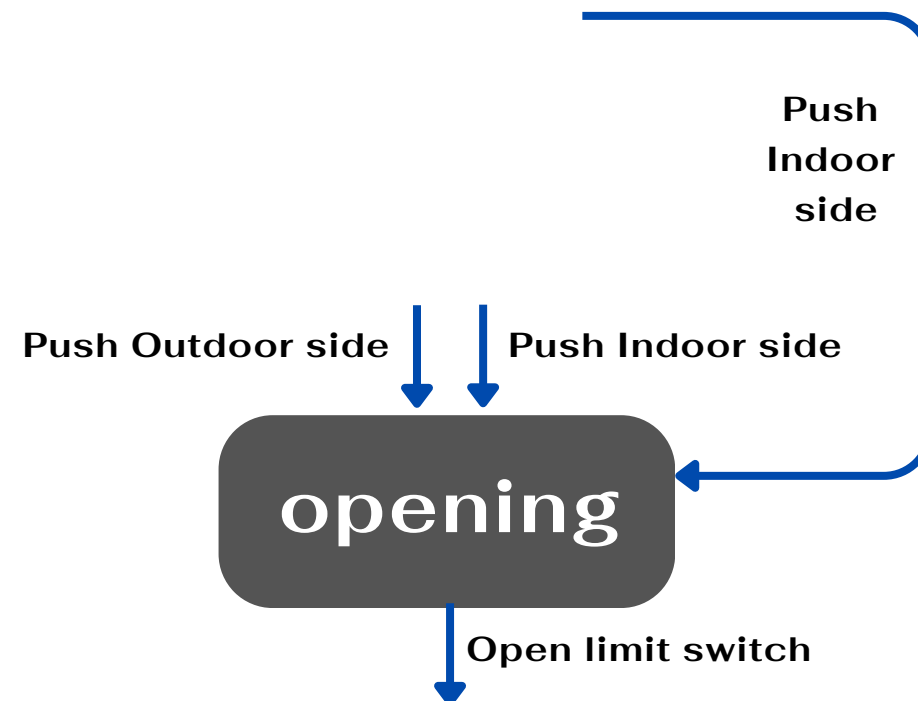
```
jB p2.0,OpeningState;open inside button
```

```
jB p2.1,OpeningState;open outside button
```

```
sjmp UnlockState
```

Code

Opening State



OpeningState:

```
mov r0,#03H ;r0 = state 3
```

```
setb p1.0 ; + pole of motor On
```

```
clr p1.1 ; - pole of motor Off
```

```
clr p1.4 ;led red Off
```

```
setb p1.5 ;led green On
```

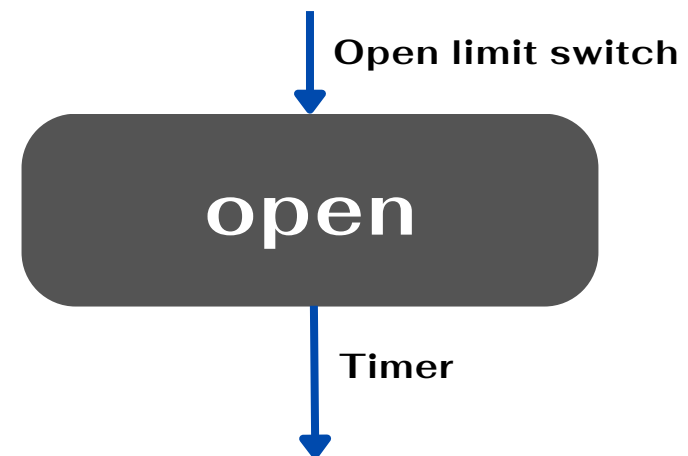
```
clr p1.7 ;bell Off
```

```
jB p2.3,OpenState ;open limit switch
```

```
sjmp OpeningState
```

Code

Open State



OpenState:

```
mov r0,#04H ;r0 = state 4
```

```
clr p1.0 ; + pole of motor Off
```

```
clr p1.1 ; - pole of motor Off
```

```
clr p1.4 ;led red Off
```

```
setb p1.5 ;led green On
```

```
clr p1.7 ;bell Off
```

```
mov r1,100 ; delay 6.5 sec
```

```
timer0:
```

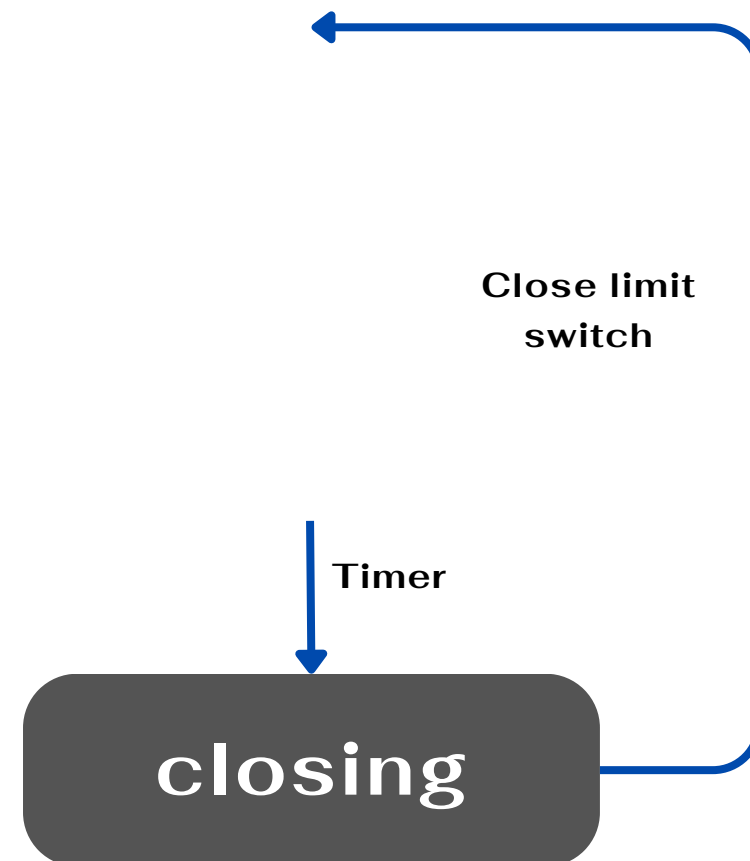
```
call delay
```

```
djnz r1,timer0
```

```
sjmp ClosingState
```

Code

Closing State



ClosingState:

```
mov r0,#05H ;r0 = state 5
```

```
clr p1.0 ; + pole of motor Off  
setb p1.1 ; - pole of motor On
```

```
clr p1.4 ;led red Off  
setb p1.5 ;led green On
```

```
clr p1.7 ;bell Off
```

```
jB p2.4,CloseState ;close limit switch
```

```
sjmp ClosingState
```

delay:

```
mov tmod,#01H ;timer0 model
```

```
mov th0,#00000000b;00H
```

```
mov tl0,#00000000b;00H delay65K usec
```

```
mov tcon,#10H ;start timer
```

```
wait: jnb tcon.5,wait
```

```
mov tcon,#00H ;stop timer
```

```
ret
```